

# Notes: Townsend (JET, 1979)

## Optimal Contracts and Competitive Markets with Costly State Verification

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# 1 Big picture

**Question.** Why do many real-world contracts look simple rather than fully state contingent? Why do firms issue debt that pays a fixed amount most of the time, and why do insurance policies often include deductibles?

**Setting.** The paper studies exchange under uncertainty when one party privately observes the realized state and the other party can learn it only by paying a verification cost. The two leading interpretations are corporate finance and insurance.

**Core friction.** State-contingent trade is useful for risk sharing, but making transfers contingent on the realized state is not free. The state has to be verified, and verification consumes resources.

**Main conceptual contribution.** Townsend shows that once one imposes incentive compatibility and costly verification, optimal contracts become simple in a very particular way: there is a region of states in which no verification occurs and a constant transfer is paid, and a low-state region in which verification occurs and the transfer can depend on the realized state.

**Main deterministic-contract result.** Under broad conditions, the optimal verification set is a *lower interval*. Low realizations are verified; high realizations are not. In the debt interpretation, low cash-flow states trigger default and audit. In the insurance interpretation, only sufficiently large losses trigger claims, so the contract behaves like a deductible policy.

**Main robustness message.** The low-state-cutoff result survives not only under a smooth convex verification-cost specification but also, with additional assumptions, under a fixed verification cost.

**Main caveat.** Deterministic verification is not always optimal. A stochastic auditing scheme can dominate the best deterministic contract because the threat of random verification may preserve incentives while saving expected resource costs.

**General-equilibrium contribution.** In a multi-agent extension with bilateral and separability restrictions, Townsend defines a competitive equilibrium with contingent claims and shows that equilibrium exists and implements a constrained optimum.

**Economic takeaway.** Costly state verification gives a disciplined explanation for why optimal risk-sharing arrangements often look like fixed debt with bankruptcy only in bad states or insurance with deductibles rather than fully contingent Arrow–Debreu contracts.

## 2 Model environment and key objects

### 2.1 Endowments, preferences, and feasibility

There are two agents and one consumption good. Agent 1 has a nonrandom endowment  $y_1 > 0$ . Agent 2 has a random endowment  $y_2 \in [\alpha, \beta]$ , where  $\alpha > 0$ . The variable  $y_2$  is either discrete with finitely many states or continuous with density  $f(y_2) > 0$  on  $[\alpha, \beta]$ .

Each agent  $j$  has Bernoulli utility  $U_j(c_j)$ , increasing and concave in riskless consumption. Townsend additionally assumes strict concavity and Inada-style conditions for agent 2:

$$U'_2(0) = \infty, \quad U'_2(\infty) = 0.$$

Feasibility requires

$$c_1(y_2) + c_2(y_2) \leq y_1 + y_2.$$

**Interpretation.** The paper starts from a standard risk-sharing environment. If information were public, the agents would want state-contingent trade. The paper asks how this changes once the state is privately observed and verification is costly.

## 2.2 Information asymmetry and costly verification

The realization of  $y_2$  is known only by agent 2 unless verification occurs. Verification reveals  $y_2$  perfectly to agent 1 but uses up resources.

Townsend allows two verification-cost specifications:

- a constant cost  $\mu > 0$ ;
- or a cost that depends on the transfer, written as  $\xi[g(y_2)]$ .

**Interpretation.** In the firm interpretation, verification is costly auditing or bankruptcy investigation. In the insurance interpretation, it is the cost of investigating or validating a claim.

## 2.3 Contract definition

A contract is a prestate agreement  $[g, S]$ , where

- $g(y_2)$  is the transfer from agent 2 to agent 1 as a function of the realized state;
- $S$  is the set of states in which verification is supposed to occur.

If  $y_2 \in S$ , agent 2 asks for verification, pays the verification cost, the state is observed, and the transfer  $g(y_2)$  is carried out.

If  $y_2 \notin S$ , there is no verification. Since agent 2 controls the announcement and prefers to transfer as little as possible, the actual transfer in the nonverification region is the smallest transfer consistent with the contract.

**Interpretation.** The paper makes explicit that unverifiable contingencies are meaningless in equilibrium. If the contract tries to condition on information that the uninformed party cannot check, the informed party will exploit that discretion.

## 2.4 Consistency: what implementable contracts look like

A contract is *consistent* if agent 2 wants to follow the prescribed verification rule and the actual transfer schedule coincides with the promised one.

The key characterization is Lemma 2.1. A contract is consistent if and only if:

$$g(y_2) = C \quad \text{for } y_2 \in S',$$

for some constant  $C$ , and

$$g(y_2) + \xi[g(y_2)] < C \quad \text{for } y_2 \in S.$$

**Interpretation.** On the no-verification set, the transfer cannot depend on the hidden state, so it collapses to a constant. On the verification set, the informed party must strictly prefer verification to pretending to be in the no-verification region. This is the paper's basic incentive-compatibility logic.

## 2.5 Why focusing on consistent contracts is without loss

Lemma 2.2 states that for any arbitrary contract  $[g, S]$ , one can construct a consistent contract  $[\tilde{g}, \tilde{S}]$  that generates the same allocation.

**Interpretation.** Moral-hazard concerns are not added on top of the model as an afterthought. The allocation process itself reduces every feasible arrangement to the implementable one implied by incentives. It is therefore enough to solve the planner's problem over consistent contracts.

## 2.6 Debt and insurance interpretations

**Debt interpretation.** View agent 2 as a firm with project return  $y_2$ . The contract resembles a bond promising a fixed payment  $C$  in normal states. Only when the firm's return is sufficiently low does it default, trigger verification, and make a lower state-contingent payment.

**Insurance interpretation.** View agent 2 as an insured individual whose loss is  $\beta - y_2$ . The constant  $C$  is a premium, and the indemnity paid to agent 2 is

$$I(y_2) = C - g(y_2).$$

When there is no verification,  $I(y_2) = 0$ . Claims are filed only in sufficiently bad states, so the contract behaves like a deductible insurance policy.

**Interpretation.** The same formal mechanism explains both defaultable debt and insurance deductibles. In each case, verification is reserved for the bad states in which contingent adjustment is most valuable.

## 3 Two-agent optimal contracts under deterministic verification

### 3.1 Problem 3.1: the planner's problem in transfers

Townsend characterizes the Pareto frontier by maximizing agent 2's expected utility subject to agent 1's participation constraint and implementability:

$$\max_{g(\cdot), C, S} \left\{ \int_S U_2(y_2 - g(y_2) - \xi[g(y_2)]) dF(y_2) + \int_{S'} U_2(y_2 - C) dF(y_2) \right\}, \quad (1)$$

subject to

$$\int_S U_1(y_1 + g(y_2)) dF(y_2) + \int_{S'} U_1(y_1 + C) dF(y_2) \geq K, \quad (2)$$

$$g(y_2) + \xi[g(y_2)] < C \quad \text{for } y_2 \in S, \quad (3)$$

and

$$y_1 + g(y_2) \geq 0 \text{ for } y_2 \in S, \quad y_1 + C \geq 0 \text{ for } y_2 \in S'. \quad (4)$$

**Interpretation.** The first constraint keeps agent 1 on or above a required utility level  $K$ , the second is the consistency requirement, and the last rules out negative consumption for agent 1. The central design problem is when it is worth paying to verify the state.

### 3.2 Insurance transformation: Problem 3.2

Define the insurance payment

$$I(y_2) = C - g(y_2).$$

Then  $I(y_2) = 0$  in the no-verification region and  $I(y_2) > 0$  where verification occurs. If verification cost is written as a function of the claim size,

$$\xi[g(y_2)] = \Psi(I(y_2)),$$

the planner's problem becomes

$$\max_{I(\cdot), C} \int_{\alpha}^{\beta} U_2(y_2 + I(y_2) - C - \Psi[I(y_2)]) dF(y_2), \quad (5)$$

subject to

$$\int_{\alpha}^{\beta} U_1(y_1 - I(y_2) + C) dF(y_2) \geq K, \quad (6)$$

$$I(y_2) \geq 0, \quad (7)$$

and

$$y_1 - I(y_2) + C \geq 0. \quad (8)$$

Townsend assumes in this smooth case that  $\Psi$  is convex, continuously differentiable,  $\Psi(0) = 0$ , and  $\Psi'(0) < 1$ .

**Interpretation.** The transformation makes the problem look like optimal insurance with endogenous verification. The claim  $I(y_2)$  is positive only in states where the parties pay the verification cost.

### 3.3 Proposition 3.1: the verification set is a lower interval

Under the above assumptions, any solution to Problem 3.2 has a verification region of the form

$$S^* = \{y_2 : y_2 < \gamma\}$$

for some cutoff  $\gamma \in [\alpha, \beta]$ .

**Interpretation.** Verification occurs only in bad states. If the state is sufficiently good, the contract simply pays the fixed amount  $C$ . Once a bad enough state is audited, all still lower states are also audited.

**Economic meaning.** This is the main result of the paper. It endogenizes a default threshold for debt and a deductible threshold for insurance.

### 3.4 Why the cutoff is intuitive

Low realizations of  $y_2$  are precisely the states in which contingent adjustment is most valuable. When  $y_2$  is high, forcing a costly audit buys little extra risk sharing relative to the simple fixed payment  $C$ . When  $y_2$  is low, the benefit from adjusting the transfer is larger.

**Interpretation.** The contract reserves expensive state verification for the states where the marginal value of insurance or bankruptcy adjustment is highest.

### 3.5 Example: how the cutoff moves with verification costs

Townsend provides an example with:

- $U_1$  linear,

- $\Psi(I) = \lambda I, 0 \leq \lambda < 1$ ,
- $y_2$  uniformly distributed on  $[\alpha, \beta]$ ,
- and agent 1 held at autarky utility.

In that case, the optimal threshold  $\gamma_\lambda$  varies with the cost parameter  $\lambda$ :

- if  $\lambda = 0$ , verification is costless, full insurance is optimal, and  $\gamma_\lambda = \beta$  so verification occurs in every state;
- as  $\lambda \rightarrow 1$ , the allocation approaches autarky and  $\gamma_\lambda \rightarrow \alpha$ , so the verification region shrinks toward the empty set.

**Interpretation.** The more expensive verification becomes, the narrower the region over which the contract conditions on the realized state.

### 3.6 Fixed verification costs: Lemma 3.1

Townsend then switches to the case of a fixed verification cost  $\mu$ . With agent 1 risk neutral and nonbinding nonnegativity constraints, Lemma 3.1 shows that agent 2's consumption is constant on the verification set  $S^*$ .

**Interpretation.** Inside the audited region, the contract fully pools agent 2 across those states. The contract uses auditing to flatten consumption where adjustment is valuable, while leaving nonaudited states on the simple linear schedule  $y_2 - C$ .

### 3.7 Proposition 3.2: the cutoff survives with fixed audit cost

Under continuous  $y_2$ , fixed verification cost  $\mu$ , risk-neutral agent 1, and a technical condition

$$\sup c_2^*(y_2) \leq y_1 + \mu - \alpha,$$

Townsend proves that the verification set still takes the form

$$S^* = \{y_2 : y_2 < \gamma\}.$$

**Interpretation.** Even with a fixed cost per audit rather than a smooth convex cost, the logic of the contract is unchanged: audit low states, do not audit high states.

### 3.8 What the paper explains in finance and insurance

The deterministic-contract results imply two familiar institutional forms:

- **Defaultable debt.** A firm promises to pay  $C^*$  in normal states and is audited only when realized return is low enough that it cannot credibly stay on the fixed-payment schedule.
- **Insurance with a deductible.** An insured individual does not file claims for small losses, but once losses exceed the deductible the claim is verified and paid.

**Interpretation.** The paper replaces ad hoc nonnegativity or “loading” assumptions from the older insurance literature with an explicit contracting friction: conditioning on the state is costly because the state must be verified.

## 4 Random verification and robustness

### 4.1 Why deterministic verification may be too restrictive

Sections 2 and 3 assume a deterministic auditing rule: the state is either verified with probability one or with probability zero. Townsend explicitly questions whether this is optimal. If the threat of verification is enough to sustain truthful behavior, then random audits may save resources.

**Interpretation.** The issue is not whether verification is useful. It is whether one really needs to verify every audited state for sure.

### 4.2 The stochastic-audit scheme

Townsend studies a two-state example with  $y_2(s) < y_2(t)$ . After observing the state, agent 2 reports either  $s$  or  $t$ . If state  $w \in \{s, t\}$  is reported, verification occurs with probability  $\pi(w)$ .

If there is no verification after reporting  $w$ , agent 2 pays  $h(w)$ . If verification occurs and the true state is  $w$  while state  $w'$  was reported, the transfer is  $d(w, w')$ . The fixed cost of verification is  $\mu$ .

Truthful reporting requires

$$\begin{aligned} & [1 - \pi(w)]U_2[y_2(w) - h(w)] + \pi(w)U_2[y_2(w) - d(w, w) - \mu] \\ & \geq [1 - \pi(w')U_2[y_2(w) - h(w')]] + \pi(w')U_2[y_2(w) - d(w, w') - \mu] \end{aligned} \quad (9)$$

for  $w, w' \in \{s, t\}$ .

**Interpretation.** A low-state agent need not be audited with certainty. It is enough that lying exposes him to a sufficiently bad lottery.

### 4.3 Main message of Section 4

Townsend shows that a stochastic verification scheme can reproduce any deterministic allocation and, in the two-state example, can strictly dominate the best deterministic procedure.

**Intuition.** Start from a deterministic contract that audits the low state and never audits the high state. Then slightly reduce the low-state audit probability. If transfers are chosen correctly, truthful reporting remains incentive compatible, but expected auditing costs fall. The saved resources make both agents better off.

**Why this matters.** The paper's famous low-state cutoff result is derived for deterministic auditing. Section 4 warns the reader not to overstate it as a fully general optimality result. Random audits can do better.

### 4.4 Economic interpretation

The stochastic scheme resembles random tax audits, occasional regulatory examinations, or spot checks inside firms. In all of these environments, certainty of verification is not necessary; credible threat can substitute for actual inspection.

**Interpretation.** Deterministic auditing is a useful benchmark, but once randomization is allowed, one obtains a sharper trade-off between incentives and verification costs.



## 5 Multi-agent extension and constrained optimal contracts

### 5.1 Two-sided private information with two agents

Townsend next considers a symmetric two-agent setting in which both  $y_1$  and  $y_2$  are privately observed. A contract is now  $[g, S_1, S_2]$ , where  $S_j$  is the verification set for agent  $j$ .

Consistency becomes much more complicated. Roughly:

- if neither agent is verified, the transfer must be a constant;
- if only one agent is verified, the transfer can depend only on that agent's observed state;
- if both are verified, richer transfers are feasible, but incentive constraints link all regions together.

**Interpretation.** Once several agents have private information, unrestricted optimal contracting becomes hard because the transfer can in principle depend on many privately observed variables and many verification patterns.

### 5.2 Why Townsend imposes extra structure

To regain tractability in the  $m$ -agent economy, Townsend imposes several exogenous restrictions:

- trade is essentially bilateral;
- the bilateral transfer between agents  $i$  and  $j$  depends at most on  $y_i$  and  $y_j$ ;
- transfers are separable across agents.

Under these restrictions, the bilateral transfer takes the form

$$g_{ij}(y_i, y_j) = I_{ij}(y_j) - I_{ji}(y_i) + C_{ij},$$

with  $I_{ij}(y_j) \geq 0$ , and the net constant terms satisfy

$$\sum_{j=1}^m C_j = 0.$$

**Interpretation.** The paper is honest that this is a constrained problem, not the unrestricted multi-agent optimum. The restrictions are there so that the low-state-audit insight can be generalized in a manageable way.

### 5.3 Verification sets in the constrained $m$ -agent problem

Let  $\sum_i I_{ij}(y_j)$  denote the total insurance payment received by agent  $j$ . Verification cost for agent  $j$  is

$$\Psi_j \left[ \sum_i I_{ij}(y_j) \right].$$

Then the verification set for agent  $j$  is

$$S_j = \left\{ y_j : \sum_i I_{ij}(y_j) - \Psi_j \left[ \sum_i I_{ij}(y_j) \right] > 0 \right\}.$$

Agent  $j$ 's consumption is therefore

$$c_j = y_j + \sum_i I_{ij}(y_j) + C_j - \sum_{i \neq j} I_{ji}(y_i) - \Psi_j \left[ \sum_i I_{ij}(y_j) \right].$$

**Interpretation.** Each agent pays a premium-like constant and receives state-contingent compensation only as a function of his own realized endowment, exactly in the spirit of the two-agent insurance transformation.

#### 5.4 Problem 5.1 and Proposition 5.1

Townsend then chooses the functions  $I_{ij}$  and constants  $C_j$  to maximize one agent's expected utility subject to the participation constraints of the others.

The main result is Proposition 5.1: for every agent  $j$ , any constrained optimum has a low-state verification region

$$S_j^* = \{y_j : y_j < \gamma_j\}$$

for some threshold  $\gamma_j$ .

**Interpretation.** The low-state-cutoff logic survives in the multi-agent environment, but only after the problem has been narrowed to a restricted class of bilateral, separable contracts.

#### 5.5 What is gained and what is lost

**What is gained.** The paper shows that costly verification still rationalizes deductible-like or default-threshold-like contracts even when many agents trade.

**What is lost.** The result is about a *constrained* optimum. Richer multilateral trades and more complicated verification-contingent transfers are ruled out by assumption, not derived from the environment.

**Interpretation.** This is one of the paper's key limitations and Townsend states it openly. The section is best read as a bridge from the two-agent insight toward general equilibrium, not as a fully general multi-agent contract theory.

## 6 Competitive equilibrium

### 6.1 Contingent claims and market structure

Townsend then asks whether the constrained optimal allocations can arise from decentralized trade. Each agent  $j$  has  $n$  possible realizations  $y_j(s_j)$ , with probabilities  $p_j(s_j)$ .

Before states are realized, agents trade:

- *contingent claims* on each agent's endowment realization;
- *unconditional claims* on the consumption good.

Let  $J_{ij}(s_j)$  be agent  $i$ 's holdings of claims contingent on state  $s_j$  of agent  $j$ , and let  $D_i$  be holdings of unconditional claims.

Trade direction is restricted:

$$J_{ii}(s_i) \geq 0, \quad J_{ij}(s_j) \leq 0 \text{ for } j \neq i.$$

**Interpretation.** Agent  $i$  buys claims on his own endowment and issues claims on the endowments of others. This mirrors the earlier bilateral-insurance structure.

## 6.2 Agent $i$ 's problem

Given prices  $q_j(s_j)$  for contingent claims and price  $r$  for unconditional claims, agent  $i$  solves

$$\max_{J_{ij}(s_j), D_i} \sum_{s_1=1}^n \cdots \sum_{s_m=1}^n \left[ \prod_{j=1}^m p_j(s_j) \right] U_i \left( y_i(s_i) + \sum_{j=1}^m J_{ij}(s_j) + D_i - \Psi_i[J_{ii}(s_i)] \right), \quad (10)$$

subject to the budget constraint

$$\sum_{j=1}^m \sum_{s_j=1}^n q_j(s_j) J_{ij}(s_j) + r D_i \leq 0, \quad (11)$$

the sign restrictions above, the requirement that insurance receipts cover verification costs,

$$J_{ii}(s_i) - \Psi_i[J_{ii}(s_i)] \geq 0,$$

and nonnegative consumption:

$$y_i(s_i) + \sum_{j=1}^m J_{ij}(s_j) + D_i - \Psi_i[J_{ii}(s_i)] \geq 0.$$

**Interpretation.** This is a decentralized version of the constrained contracting problem. The contingent claim  $J_{ii}(s_i)$  is what agent  $i$  collects after state  $s_i$  occurs, and claiming it triggers verification costs.

## 6.3 Competitive equilibrium definition

A competitive equilibrium is a collection of nonnegative prices  $q_j^*(s_j)$  and  $r^*$ , together with holdings  $J_{ij}^*(s_j)$  and  $D_i^*$ , such that:

1. each agent solves his utility-maximization problem given prices;
2. markets clear:

$$\sum_{i=1}^m J_{ij}^*(s_j) \leq 0, \quad \sum_{i=1}^m D_i^* \leq 0.$$

**Interpretation.** The model is still unusual because claims are not anonymous. A claim contingent on  $y_j$  is associated with agent  $j$ , so there are many sellers but only one natural buyer.

## 6.4 Propositions 6.1 and 6.2

**Existence.** Proposition 6.1 establishes that a competitive equilibrium exists under the maintained assumptions.

**Welfare.** Proposition 6.2 shows that the allocation of any competitive equilibrium is a constrained optimum.

**Interpretation.** Within the paper's restricted class of contracts, decentralized markets reproduce the planner's constrained-efficient allocation.

## 6.5 Pareto satisfactory relative to constrained optima

Townsend argues that the equilibrium concept is Pareto satisfactory relative to the constrained optimum because:

- every equilibrium allocation is constrained optimal;
- any constrained-optimal allocation can be supported as an equilibrium if agents are endowed with the associated claims.

**Interpretation.** This is the paper's decentralization theorem. Even with moral hazard and costly verification, the restricted market structure can implement the same allocations as the planner.

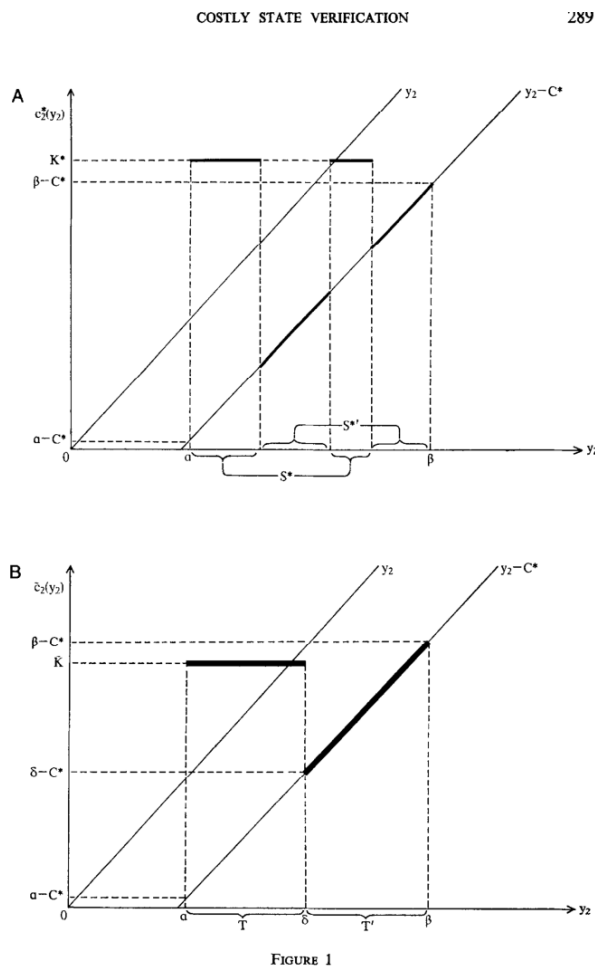
## 6.6 An important caveat about price taking

Townsend also notes a conceptual weakness. Claims are linked to specific agents, so the buyer of a claim on  $y_j$  is essentially agent  $j$  himself. This makes the assumption that agents take prices as given less innocent than in standard anonymous contingent-commodity markets.

**Interpretation.** The decentralization result is therefore best viewed as a first pass. Townsend suggests that replication and endogenous price-setting games would be needed for a fuller foundation.

## 7 Figures and key theoretical results

### 7.1 Figure 1: why a scattered audit set cannot be optimal



**Figure 1:** Appendix proof intuition for Proposition 3.2. Townsend compares a non-lower-interval audit set  $S^*$  with a lower interval  $T$  of the same probability mass.

#### Read:

- Panel A depicts an original consumption path for agent 2 in which audited states are scattered rather than concentrated at the bottom.
- Panel B constructs a new allocation with the same expected audit cost and the same mean consumption but with auditing concentrated on the lowest states.
- The thick horizontal segment shows pooling inside the audited region; the linear segment is the no-verification schedule  $y_2 - C^*$ .

**Economic message.** If the audit set is not a lower interval, one can shift auditing toward worse states and make the informed agent better off without hurting the uninformed agent. That is the contradiction behind Proposition 3.2.

## 7.2 Figure 2: second-order-stochastic-dominance logic

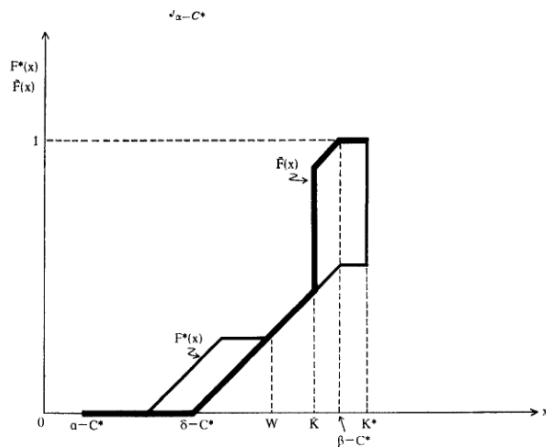


FIGURE 2

**Figure 2:** Appendix proof device for Proposition 3.2. The alternative consumption distribution  $\tilde{F}$  lies below  $F^*$  over the relevant range before catching up at the top.

### Read:

- The figure compares the cumulative distribution of the original consumption plan with that of the alternative lower-interval-audit plan.
- The alternative plan preserves the mean but shifts probability mass in a way preferred by a strictly concave utility function.
- This is the Rothschild–Stiglitz second-order-stochastic-dominance step used in the proof.

**Economic message.** The paper’s low-state-audit result is not only intuitive; it can be formalized through a clean distributional-comparison argument.

### 7.3 Summary of the main lemmas and propositions

| Result               | Formal content                                                                                                                         | Economic meaning                                                                                       |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Lemma 2.1            | Implementable contracts pay a constant $C$ on the no-verification set and must make verification strictly attractive in the audit set. | Hidden states cannot support arbitrary contingencies; unverifiable regions collapse to fixed payments. |
| Lemma 2.2            | Any contract is equivalent to a consistent one generating the same allocation.                                                         | It is enough to solve the planner's problem over implementable contracts.                              |
| Proposition 3.1      | With smooth convex verification costs, $S^* = \{y_2 < \gamma\}$ .                                                                      | Optimal deterministic contracts audit only bad states.                                                 |
| Lemma 3.1            | Under fixed audit cost and risk-neutral agent 1, agent 2 is fully pooled inside the audit set.                                         | Audited states are used to flatten agent 2's consumption.                                              |
| Proposition 3.2      | With fixed audit cost and additional assumptions, $S^* = \{y_2 < \gamma\}$ .                                                           | The deductible/default-threshold logic survives beyond the smooth-cost case.                           |
| Section 4 result     | A stochastic audit scheme can dominate the optimal deterministic scheme.                                                               | Random auditing can save expected verification costs while preserving truthful revelation.             |
| Proposition 5.1      | In the constrained $m$ -agent problem, each agent has $S_j^* = \{y_j < \gamma_j\}$ .                                                   | The low-state-cutoff logic extends to multi-agent bilateral insurance structures.                      |
| Propositions 6.1–6.2 | Competitive equilibrium exists and any equilibrium allocation is a constrained optimum.                                                | Decentralized trade can implement constrained-efficient CSV allocations.                               |

**Table 1:** Main theoretical results and their economic interpretation.

## 8 Short summary

- State-contingent trade is valuable for risk sharing, but it requires verification when only one side observes the realized state.
- Implementability forces nonverified states to share a constant transfer  $C$ , while verified states can have contingent transfers.
- In the two-agent deterministic problem, optimal verification occurs only in low states. This delivers debt with default in bad states and insurance with a deductible.
- The low-state-cutoff result survives under both smooth convex verification costs and, with extra assumptions, fixed verification costs.
- Allowing random audits weakens the claim that deterministic verification is fully optimal: stochastic auditing can dominate.
- In a restricted  $m$ -agent environment, each agent again has a low-state audit set, and a competitive equilibrium exists that implements the constrained optimum.

## 9 Conclusion

### 9.1 Core conclusion

Townsend's central message is that simple financial contracts can be an optimal response to informational frictions. Once the state is privately observed and costly to verify, the optimal contract does not fully insure every contingency. Instead, it economizes on verification and uses contingent transfers only in sufficiently bad states.

### 9.2 Economic intuition

The contract trades off two objectives:

- better risk sharing through state-contingent transfers;
- lower resource waste by avoiding unnecessary verification.

High states are handled cheaply by a fixed transfer. Low states are important enough to justify auditing. This naturally generates default thresholds and deductibles.

### 9.3 What the paper does and does not deliver

The paper succeeds in giving a clean economic reason for why debt and insurance often look simple. It also takes important steps toward competitive equilibrium with asymmetric information and costly verification.

At the same time, Townsend is explicit about the limitations. Random auditing can dominate deterministic verification, and the multi-agent equilibrium section relies on exogenous bilateral and separability restrictions. The paper is therefore best read as a foundational benchmark: it isolates the costly-state-verification mechanism clearly, even though later work is needed to generalize it.

### 9.4 Takeaway

The paper's lasting insight is that *noncontingent-looking contracts can be the optimal outcome of a deeper contingent contracting problem once verification itself is costly*. That insight is what makes the fixed payment of debt and the deductible of insurance economically intelligible rather than ad hoc.